Acknowledgements

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Contents

Chapter 1

Introduction

1.1 Motivation

Financial time series data is never fully observed Limit-order book feeds drop quotes during bursts of order cancellations. Equity markets suspend trading when prices move too violently. Over-the-counter bond markets leave entire yield-curve tenors unquoted for days. Implied volatility surfaces contain large regions where no liquid option trade has occurred. Macro and alternative data arrive at mixed, irregular frequencies, with publication lags that create structural gaps at the daily level. This missing data is pervasive and often elusive, and gets in the way of advanced analysis. Thus practitioners or a researchers face the same decision: what value should be attributed to the missing observation, and how much uncertainty should be attached to that attribution?

The simplest and most common solution is to carry forward the last observed value, but this can lead to systematic mispricing, underestimation of risk, and distortion of correlations among assets.

We aim to take a different path, one that looks for solutions that aim to preserve as much as possible the statistical properties of the series, in order to maintain the integrity of the underlying data and consequently the robustness of any downstream model that relies on it. Since this can be considered an inference problem, we will also explore ML algorithms, and in classical machine learning approach, we will focus on quality of inference rather than interpretability of the model.

In this thesis we will test these different methods of imputation, and evaluate them not only on the accuracy of the imputation itself (MSE,...), but also on the quality of the downstream model that uses the imputed data, and how far it deviates from the case where all data is observed.

1.2 Literature review

1.3 Research questions

1.4 Contributions

1.5 Outline of the thesis

Chapter 2

Methodology

As already mentioned, missing data can arise in countless different ways, and the choice of methodology depends on the nature of the missingness, the type of data, and the specific goals of the analysis. We will focus on a subset of cases, namely those that involve imputation of the missing values with "information" from the whole dataset (no causal constraint), with in mind the goal of training a downstream model that is as close as possible to the one that would have been trained on the complete dataset. Our choice of data, models, and evaluation methods will be guided by these constraints. We will nevertheless mention when/how models should be tweaked to solve adjacent problems.

2.1 Data

As our main baseline, we chose the most studied and reliable dataset in finance: the end-of-day prices of the S&P500, in this case fetched with yfinance from 1/1/2010 to 30/5/2026 (potrei cambiare le date). It clearly is not a dataset inflicted with the curse of missing data; the only structural ones being occasional circuit breaks and listings/delistings from the index. To obtain a full dataset with no "real" missing data, we removed each stock that contained at least one missing observation. Furthermore, we selected a sample of 50 stocks from the index, stratified across industries; this is a sensible choice because:

1. with more 400 stocks the covariance matrix $\mathbf{\Sigma}$ quickly becomes unreliable (noisy covariances, skewed eigenvalues, unstable inversion. . .) as the number of observations shrinks;

2. it is easier to check whether the ground data is actually correct;

3. model training requires considerably less memory, in a time where RAM and GPU prices are skyrocketing;

4. it is well known that a well constructed portfolio with more than 30 stocks is already sufficiently diversified (since volatility scales with the inverse of the square root of

the number of securities).

This dataset will serve as the ground truth against which we will compare our results. In parallel, we will modify it by inserting fake missing observations according to different rules, as we define in the subsequent sections.

Now an important choice is whether to operate in the prices "space" P_t or the log returns space R_t. There are positives and negatives about each. For one, the raw dataset is prices, and in practice missing data will mostly come in the form of missing prices. Furthermore, in most applications we care slightly more about prices than returns, since the formers are what is actually bought and sold. But P_t has the drawback of not being a complete space ($p_{t,k} > 0$) and being highly level dependant, making the series non-stationary and very difficult for ML algorithms to handle. We will thus focus on R_t.

2.1.1 Missingness Injection

(Da scrivere meglio, questa è solo l'outline)

As outlined in the literature review ...

We used three algorithms to simulate different missingness patterns:

1. **Sparse**: each scalar $r_{t,k}$ is missing with probability r, independent of the rest of the dataset. This is the easiest example of MCAR and can simulate a randomly corrupted dataset;

2. **Contiguous gaps**: a point $r_{t,k}$ is chosen at random and eliminated. Then subsequent observations of the same variable $r_{t+1,k}$, ..., $r_{t+h,k}$ get deleted according to an exponential decay with rate λ. This is still an example of MCAR and can be thought to simulate assets failing to trade because of random shortages of liquidity;

3. **Stressed**: we calculate index squared returns for each date t and rank them. Then, returns whose date is above a certain percentile get deleted with probability h, while the rest still go missing with rate r. This simulates shortages of liquidity or circuit breaks that happen when markets experience high levels of volatility; since the missingness of a given $r_{t,k}$ is injected conditional on all the other values of the dataset, this is an example of MNAR.

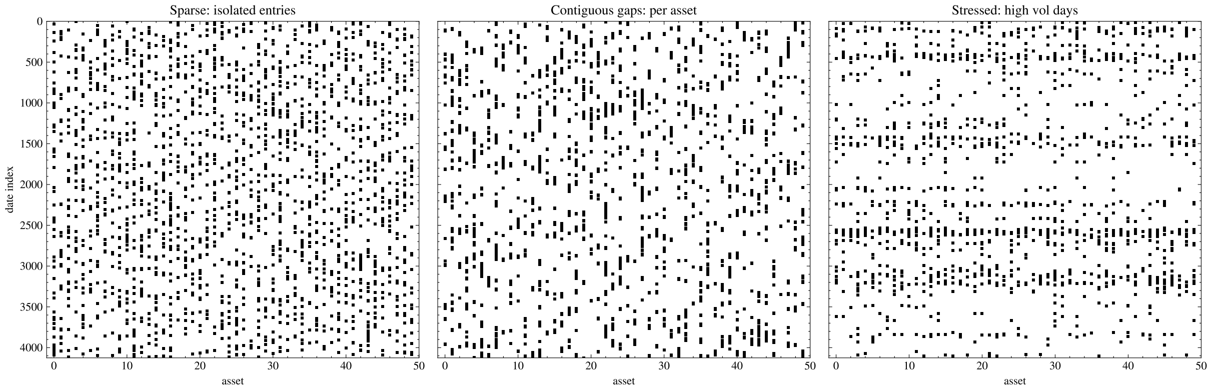

Figure 2.1: Missingness Examples

2.2 Models

2.2.1 Deletion

Deletes rows where at least one missing value is found

2.2.2 Inserting zeroes

Consists in setting each missing return to zero. this is equivalent to setting the missing price to be equal to the last observed price.

2.2.3 EM

With student-t

Dempster et al. (1977)

2.2.4 KNN

2.2.5 BRITS

Cao et al. (2018)

2.2.6 CSDI

2.3 Evaluation Methods

MSE

MAEa

CovErr =

$$\frac{\|\widehat{\Sigma}_{\text{imp}} - \Sigma_{\text{true}}\|}{\|\Sigma_{\text{true}}\|}$$

VaR

Minimum Variance Portfolio

2.4 Geometrical Interpretation

In this section we want to develop some geometrical intuition behind the evaluation methods criteria and the imputation models.

First consider a full multivariate time series of daily log returns:

$$R = \begin{pmatrix} r_1^\top \\ r_2^\top \\ \vdots \\ r_T^\top \end{pmatrix} = \begin{pmatrix} r_{11} & r_{12} & \cdots & r_{1N} \\ r_{21} & r_{22} & \cdots & r_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ r_{T1} & r_{T2} & \cdots & r_{TN} \end{pmatrix} \in \mathbb{R}^{T\times N}$$

The (t)-th observation is:

$$r_t = (r_{t1}, \ldots, r_{tN})^\top \in \mathbb{R}^N$$

The (i)-th variable asset realization is:

$$R_i = (r_{1i}, \ldots, x_{Ti})^\top \in \mathbb{R}^T$$

Let us first consider each asset separately. We assume each return series is a realization of a square integrable random value in a Hilbert space:

$$R_i \in L^2(\Omega, \mathcal{F}, \mathbb{P}) = \left\{X : \Omega \to \mathbb{R} : \mathbb{E}[X^2] < \infty\right\}$$

Square integrability, which is a reasonable assumption [1], enables us to define an useful inner product on L^2, namely:

$$\langle X, Y\rangle = \mathbb{E}[XY]$$

Now this is remarkably similar to the definition of covariance, and is equal when the mean of X and Y is zero; at the daily level this is almost the case.

[1] Empirical work as Plerou et al. (1999) shows that the power law exponent of individual stocks is around 3, meaning variance exists but skew and kurtosis may not.

A multivariate time series is represented as

Hilbert space

The Hilbert space of square-integrable random variables is

Inner product

Norm

$$\|X\| = \sqrt{\langle X, X\rangle} = \sqrt{\mathbb{E}[X^2]}.$$

Distance

$$d(X,Y) = \|X - Y\|.$$

Centered random variables

If

$$\mathbb{E}[X] = 0, \qquad \mathbb{E}[Y] = 0,$$

then

$$\langle X, Y\rangle = \mathrm{Cov}(X,Y).$$

Correlation

$$\rho(X,Y) = \frac{\langle X, Y\rangle}{\|X\|\|Y\|}.$$

Angle between random variables

$$\theta = \arccos\left(\frac{\langle X, Y\rangle}{\|X\|\|Y\|}\right).$$

Conditional expectation

Conditional expectation

$$\mathbb{E}[X \mid \mathcal{G}].$$

Projection interpretation

$$\mathbb{E}[X \mid \mathcal{G}] = P_{L^2(\mathcal{G})}(X),$$

where ($P_{L^2(\mathcal{G})}$) denotes the orthogonal projection onto the closed subspace ($L^2(\mathcal{G})$).

Orthogonality condition

$$X - \mathbb{E}[X \mid \mathcal{G}] \perp L^2(\mathcal{G}).$$

Equivalently,

$$\mathbb{E}\left[(X - \mathbb{E}[X \mid \mathcal{G}])\, Y\right] = 0, \qquad \forall Y \in L^2(\mathcal{G}).$$

Covariance matrix

Mean vector

$$\mu = \mathbb{E}[X].$$

Covariance matrix

$$\Sigma = \mathbb{E}\left[(X - \mu)(X - \mu)^\top\right].$$

Sample covariance

$$S = \frac{1}{T-1} \sum_{t=1}^{T} (x_t - \bar{x})(x_t - \bar{x})^\top.$$

Correlation matrix

$$R = D^{-1/2} \Sigma D^{-1/2},$$

where

$$D = \operatorname{diag}(\Sigma).$$

Gram matrix

The Gram matrix of variables

$$G = X^\top X.$$

The Gram matrix of observations

$$K = X X^\top.$$

PCA

Eigenvalue decomposition

$$\Sigma = Q \Lambda Q^\top.$$

Projection onto first (k) principal directions

$$\widehat{X} = X Q_k Q_k^\top.$$

Low-rank approximation

$$X \approx UV^\top.$$

Mahalanobis geometry

Mahalanobis distance

$$d_M(x, \mu) = \sqrt{(x-\mu)^\top \Sigma^{-1} (x-\mu)}.$$

Squared Mahalanobis distance

$$\delta = (x-\mu)^\top \Sigma^{-1} (x-\mu).$$

Ellipsoid

$$\mathcal{E} = \{x : (x-\mu)^\top \Sigma^{-1} (x-\mu) \le c\}.$$

Partitioned covariance matrix

$$\Sigma = \begin{pmatrix} \Sigma_{oo} & \Sigma_{om} \\ \Sigma_{mo} & \Sigma_{mm} \end{pmatrix}.$$

Conditional mean

$$\mu_{m|o} = \mu_m + \Sigma_{mo} \Sigma_{oo}^{-1} (x_o - \mu_o).$$

Conditional covariance

$$\Sigma_{m|o} = \Sigma_{mm} - \Sigma_{mo} \Sigma_{oo}^{-1} \Sigma_{om}.$$

Projection matrices

Orthogonal projector

$$P = P^\top = P^2.$$

Least-squares projection

$$P = X(X^\top X)^{-1} X^\top.$$

Projected vector

$$\widehat{y} = Py.$$

Residual

$$r = (I - P)y.$$

Orthogonality

$$P(I-P) = 0.$$

Matrix norms

Frobenius norm

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2}.$$

Spectral norm

$$\|A\|_2 = \sigma_{\max}(A).$$

Nuclear norm

$$\|A\|_* = \sum_i \sigma_i(A).$$

Matrix completion

Projection operator

$$P_\Omega(X).$$

Optimization problem

$$\min_M \frac{1}{2}\|P_\Omega(X-M)\|_F^2 + \lambda\|M\|_*.$$

Student-(t) distribution

$$X \sim t_\nu(\mu, \Sigma).$$

Student-(t) weight

$$w_i = \frac{\nu + p}{\nu + \delta_i}.$$

EM algorithm

Expected complete-data log-likelihood

$$Q(\theta|\theta^{(k)}) = \mathbb{E}\left[\log p(X_{\text{obs}}, X_{\text{mis}} \mid \theta) \mid X_{\text{obs}}, \theta^{(k)}\right].$$

M-step

$$\theta^{(k+1)} = \arg\max_\theta Q(\theta|\theta^{(k)}).$$

Random paths

A multivariate path

$$X : [0, T] \to \mathbb{R}^N.$$

Time-parametrized market trajectory

$$t \mapsto X_t.$$

Signatures

Signature

$$S(X) = \left(1, \int dX, \int dX \otimes dX, \int dX \otimes dX \otimes dX, \dots\right).$$

Truncated signature

$$S^{(m)}(X).$$

Signature distance

$$d_{\text{sig}} = \left\| S^{(m)}(X) - S^{(m)}(\widehat{X}) \right\|.$$

Useful notation

Expectation

$$\mathbb{E}[X].$$

Probability

$$\mathbb{P}(A).$$

Variance

$$\mathrm{Var}(X).$$

Covariance

$$\mathrm{Cov}(X, Y).$$

Trace

$$\mathrm{tr}(A).$$

Determinant

$$\det(A).$$

Rank

$$\mathrm{rank}(A).$$

Diagonal matrix

$$\mathrm{diag}(a_1, \ldots, a_n).$$

Identity matrix

$$I_n.$$

Positive-definite matrices

$$\mathcal{S}^n_{++}.$$

Symmetric matrices

$$\mathcal{S}^n.$$

Chapter 3

Imputation Methods

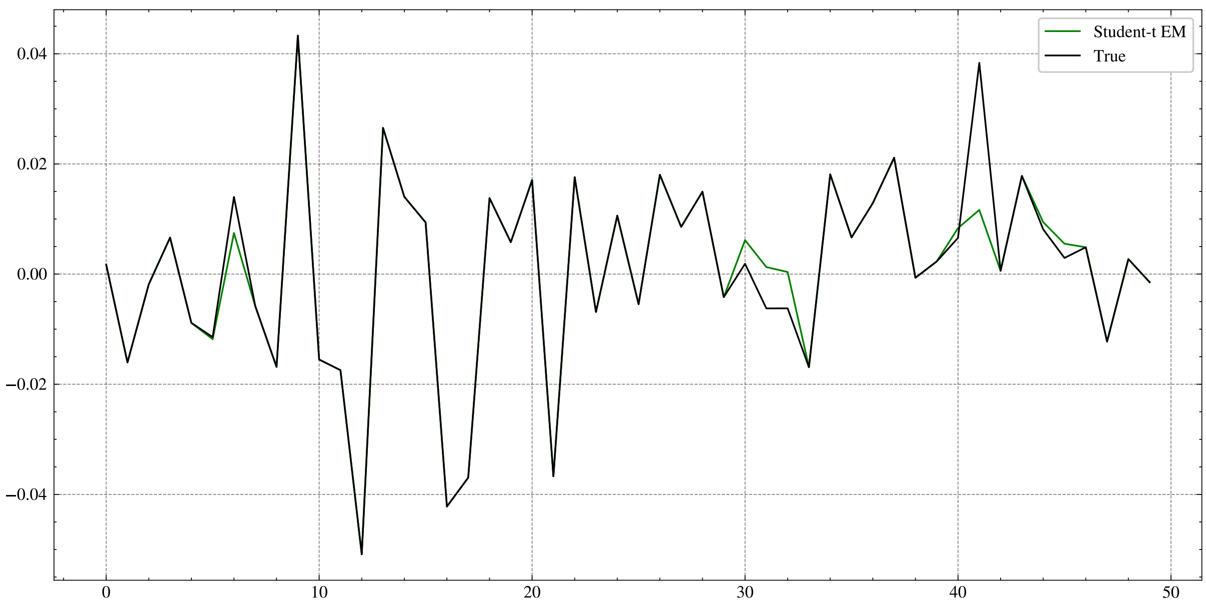

Figure 3.1: Examples of Imputed Data

Chapter 4

Results and Discussion

Chapter 5

Conclusion

5.1 Summary

5.2 Limitations

5.3 Future Work

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